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School: ..... Campus: .....

Academic Year: ..... Subject Name: ..... Subject Code: .....

Semester: ..... Program: ..... Branch: ..... Specialization: .....

Date: .....

Applied and Action Learning
(Learning by Doing and Discovery)

Name of the Experiment:- Clean a raw dataset ,handle missing values and Perform Feature Engineering

Coding Phase:-Pseudo Code / Flow Chart / Algorithm

BEGIN

Step 1: Import Required Libraries
- Import pandas
- Import numpy
- Import seaborn
- Import matplotlib

Step 2: Create the Dataset
- Create a dictionary containing:
  - Name
  - Age
  - Gender
  - Marks
  - City
- Convert the dictionary into a DataFrame

Step 3: Check Missing Values
- Display the number of missing values in each column

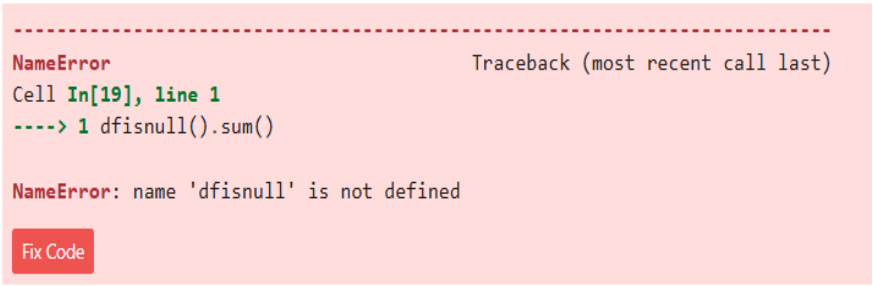
Step 4: Handle Missing Values
- Replace missing values in Age with the median of Age
- Replace missing values in Marks with the median of Marks
- Replace missing values in City with the mode of City
- Display the updated missing value count

Step 5: Perform Feature Engineering
- Create Age_Group
  - Divide Age into categories:
    - Child
    - Young
    - Adult
    - Middle Age
    - Senior
- Create Performance
  - Divide Marks into categories:
    - Fail
    - Pass
    - Good
    - Very Good
    - Excellent

Step 6: Display Results
- Display Age and Age_Group
- Display Marks and Performance

END

Testing Phase:-Compilation of Code (error detection)



Implementation Phase :- Final Output(no error)

Step 1: Import Required Libraries

```
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Create / Load Dataset

```
data = {
  "Name": [
    "Amit", "Priya", "Rahul", "Sneha", "Rohan",
    "Anjali", "Vikash", "Neha", "Suman", "Karan",
    "Pooja", "Arjun", "Meena", "Rakesh", "Nisha",
    "Deepak", "Kavita", "Ajay", "Simran", "Ravi"
  ],

  "Age": [
    25, 30, np.nan, 28, 35,
    22, 29, np.nan, 31, 27,
    26, 34, 24, 32, np.nan,
    29, 23, 36, 28, 30
  ],

  "Gender": [
    "Male", "Female", "Male", "Female", "Male",
    "Female", "Male", "Female", "Female", "Male",
    "Female", "Male", "Female", "Male", "Female",
    "Male", "Female", "Male", "Female", "Male"
  ],

  "Salary": [
    40000, 50000, 45000, np.nan, 60000,
    38000, 52000, 48000, np.nan, 47000,
    43000, 62000, 39000, 58000, 45000,
    np.nan, 41000, 65000, 49000, 53000
  ],
```

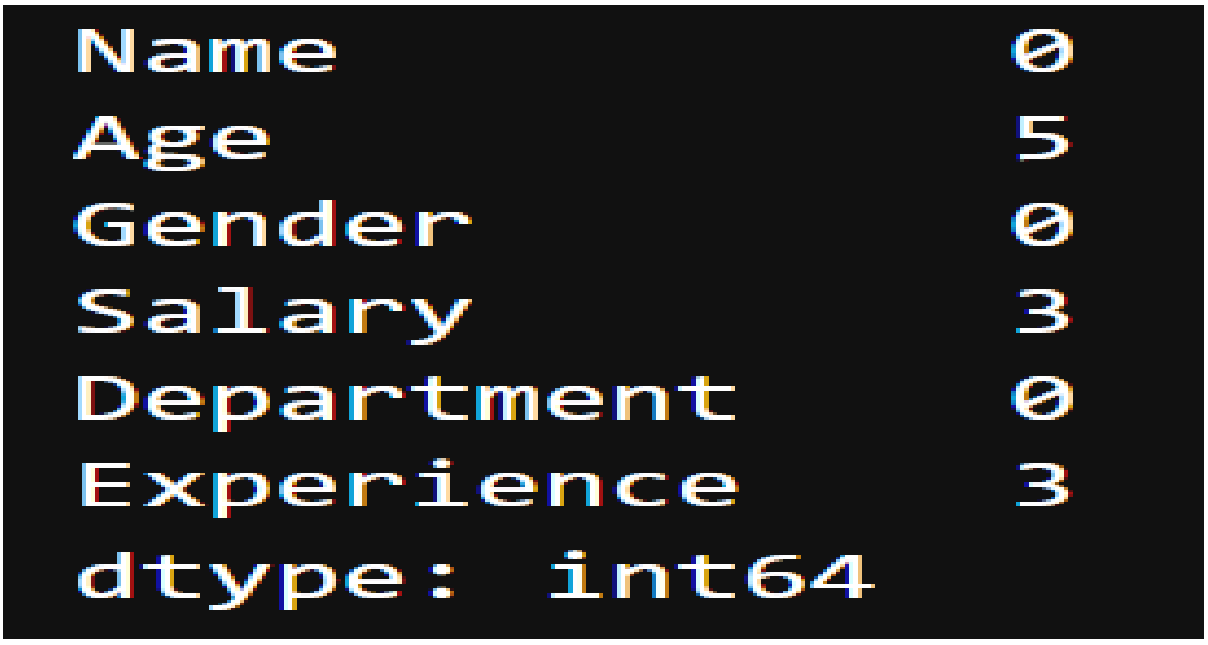
```
    "Department": [
        "HR", "IT", "Sales", "Finance", "IT",
        "HR", "Sales", "Finance", "IT", "HR",
        "Sales", "IT", "Finance", "HR", "Sales",
        "IT", "Finance", "HR", "Sales", "IT"
    ],

    "Experience": [
        2, 5, 3, 4, np.nan,
        1, 6, 2, 5, 3,
        np.nan, 7, 2, 6, 4,
        5, 1, 8, np.nan, 4
    ]
}

# Convert into DataFrame
df = pd.DataFrame(data)
```

Step 3: Check Missing Value

```
df.isnull().sum()
```



Step 4: Handle missing

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
df["Experience"] = df["Experience"].fillna(df["Experience"].mean())
print(df.isnull().sum())
```

```
Name          0
Age           0
Gender        0
Salary        0
Department    0
Experience    0
dtype: int64
```

Step 5: **Perform Feature Engineering**

Create Age Group

```
# Create Age Group

df["Age_Group"] = pd.cut(
    df["Age"],
    bins=[0,18,25,40,130],
    labels=["Child","Young","Adult","Senior"]
)

print(df[["Age","Age_Group"]].head())
```

```
    Age Age_Group
0  25.0     Young
1  30.0     Adult
2   NaN       NaN
3  28.0     Adult
4  35.0     Adult
```

Create Performance Column

```
# Create Performance Column

df["Performance"] = pd.cut(
    df["Salary"],
    bins=[0,40000,50000,60000,70000],
    labels=["Low","Average","Good","Excellent"]
)

print(df[["Salary","Performance"]].head())
```

```
    Salary Performance
0  40000.0         Low
1  50000.0     Average
2  45000.0     Average
3      NaN         NaN
4  60000.0        Good
```

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	**50**		

Signature of the Student:

Name :

Reg. NO :

Signature of the Faculty: